Light

GCSE Computer Science - Programming Fundamentals (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset ← Back to Programming Fundamentals

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the real data type used for?

- ○ **A.** Storing numbers that can have decimal points (fractional values)
- ○ **B.** Storing whole numbers without decimal points
- ○ **C.** Storing only True or False values
- ○ **D.** Storing sequences of text characters

Q2 (1 mark)

What is meant by 'sequence' in programming?

- ○ **A.** A loop that repeats code multiple times

- ○ **B.** A decision point where different paths can be taken
- ○ **C.** Executing instructions one after another in the order they are written
- ○ **D.** A named block of reusable code

Q3 (1 mark)

What is a global variable?

- ○ **A.** A variable that only exists inside one subroutine
- ○ **B.** A variable that cannot be changed after declaration
- ○ **C.** A variable that stores geographical location data
- ○ **D.** A variable that is declared outside all subroutines and can be accessed from anywhere in the program

Q4 (1 mark)

What does the relational operator >= check?

- ○ **A.** If two values are exactly equal
- ○ **B.** If the left value is less than the right value
- ○ **C.** If the left value is not equal to the right value
- ○ **D.** If the left value is greater than or equal to the right value

Q5 (1 mark)

Where is the condition checked in a REPEAT-UNTIL loop?

- ○ **A.** At the end of each iteration, after the loop body has executed
- ○ **B.** At the start of each iteration, before the loop body executes
- ○ **C.** Only once at the very beginning of the loop
- ○ **D.** The condition is never checked in this type of loop

Q6 (1 mark)

In most programming languages, which arithmetic operation is performed first when no brackets are used?

- ○ **A.** Addition
- ○ **B.** Multiplication and division (before addition and subtraction)
- ○ **C.** Subtraction
- ○ **D.** Operations are always performed left to right regardless of type

Q7 (1 mark)

When does the Boolean operator OR return True?

- ○ **A.** Only when both operands are True
- ○ **B.** Only when both operands are False
- ○ **C.** When at least one operand is True

- ○ **D.** Only when exactly one operand is True

Q8 (1 mark)

What symbol is commonly used to concatenate (join) strings in many programming languages?

- ○ **A.** The minus sign (-)
- ○ **B.** The forward slash (/)
- ○ **C.** The asterisk (*)
- ○ **D.** The plus sign (+)

Q9 (1 mark)

What is debugging?

- ○ **A.** The process of finding and fixing errors in a program
- ○ **B.** Adding new features to a program
- ○ **C.** Converting code from one programming language to another
- ○ **D.** Making a program run faster

Q10 (1 mark)

What is one key advantage of using subroutines in a program?

- ○ **A.** Subroutines make programs use more memory

- ○ **B.** Subroutines force the program to run more slowly
- ○ **C.** Subroutines allow code to be written once and reused multiple times, reducing repetition
- ○ **D.** Subroutines prevent any errors from occurring in the program

Q11 (1 mark)

Why is input validation important in programming?

- ○ **A.** To ensure that data entered by users is appropriate and will not cause errors or unexpected behaviour
- ○ **B.** To make the program's interface more colourful
- ○ **C.** To speed up program execution
- ○ **D.** To reduce the file size of the program

Q12 (1 mark)

What is the purpose of an OUTPUT statement in programming?

- ○ **A.** To receive data from the user
- ○ **B.** To perform calculations on variables
- ○ **C.** To display information on the screen or send data to an external destination
- ○ **D.** To store values in variables

Q13 (2 marks)

What are the values of a and b after this code executes?

```
a <- 5
b <- 3
temp <- a
a <- b
b <- temp
```

- ○ **A.** a = 5, b = 3
- ○ **B.** a = 3, b = 3
- ○ **C.** a = 3, b = 5
- ○ **D.** a = 5, b = 5

Q14 (2 marks)

What is output when `x = 15` and `y = 20`?

```
IF x > 10 THEN
    IF y > 25 THEN
        OUTPUT "Both high"
    ELSE
        OUTPUT "X high only"
    ENDIF
ELSE
    OUTPUT "X not high"
ENDIF
```

- ○ **A.** Both high
- ○ **B.** Nothing is output
- ○ **C.** X not high

- **D.** X high only

Q15 (2 marks)

How many times does the word "Hello" get output by this code?

```
FOR i <- 1 TO 5
    IF i MOD 2 = 1 THEN
        OUTPUT "Hello"
    ENDIF
ENDFOR
```

- **A.** 2
- **B.** 3
- **C.** 5
- **D.** 0

Q16 (2 marks)

Given `word <- "PYTHON"`, what is the value of `SUBSTRING(word, 2, 3)` if SUBSTRING takes (string, startPosition, length) with zero-based indexing?

- **A.** PYT
- **B.** YTH
- **C.** THO
- **D.** HON

Q17 (2 marks)

What is the value of `result` after this code executes?

```
a <- 20
b <- 6
result <- a DIV b + a MOD b
```

- ○ **A.** 5
- ○ **B.** 3
- ○ **C.** 8
- ○ **D.** 26

Q18 (2 marks)

If `loggedIn = False`, what is the value of `NOT loggedIn`?

- ○ **A.** False
- ○ **B.** True
- ○ **C.** 0
- ○ **D.** Error

Q19 (2 marks)

This code finds the maximum value in an array. What is output?

```
numbers <- [7, 2, 9, 4, 6]
max <- numbers[0]
FOR i <- 1 TO 4
```

```
    IF numbers[i] > max THEN
        max <- numbers[i]
    ENDIF
ENDFOR
OUTPUT max
```

- A. 7
- B. 2
- C. 9
- D. 6

Q20 (2 marks)

What is output by this code?

```
FUNCTION square(n)
    RETURN n * n
ENDFUNCTION

result <- square(4) + square(3)
OUTPUT result
```

- A. 7
- B. 12
- C. 25
- D. 49

Q21 (2 marks)

What is the purpose of this code structure?

```
OUTPUT "Enter a positive number"
number <- USERINPUT
WHILE number <= 0
    OUTPUT "Invalid. Enter a positive number"
    number <- USERINPUT
ENDWHILE
```

- ○ **A.** To keep prompting until the user enters a positive number, rejecting zero and negative values
- ○ **B.** To count how many numbers the user enters
- ○ **C.** To add all the numbers the user enters
- ○ **D.** To display positive numbers only

Q22 (2 marks)

Given this record definition and variable, which statement correctly accesses the student's age?

```
RECORD Student
    name: String
    age: Integer
    grade: String
ENDRECORD

pupil <- Student("Emma", 15, "A")
```

- ○ **A.** age.pupil
- ○ **B.** Student.age
- ○ **C.** pupil[age]
- ○ **D.** pupil.age

Q23 (2 marks)

What is output by this code?

```
x <- 10

SUBROUTINE changeX()
    x <- 5
    OUTPUT x
ENDSUBROUTINE

changeX()
OUTPUT x
```

- ○ **A.** 5 then 5
- ○ **B.** 5 then 10
- ○ **C.** 10 then 5
- ○ **D.** 10 then 10

Q24 (3 marks)

What is output by this code?

```
FUNCTION add(a, b)
    RETURN a + b
ENDFUNCTION

FUNCTION multiply(a, b)
    RETURN a * b
ENDFUNCTION

x <- add(3, 4)
y <- multiply(x, 2)
z <- add(y, x)
OUTPUT z
```

- A. 14
- B. 28
- C. 17
- D. 21

Q25 (3 marks)

How many times is the character '*' output by this code?

```
FOR row <- 1 TO 3
    FOR col <- 1 TO row
        OUTPUT "*"
    ENDFOR
ENDFOR
```

- A. 3
- B. 6
- C. 9
- D. 12

Q26 (3 marks)

What is output by this code?

```
scores <- [80, 90, 70, 100]
total <- 0
FOR i <- 0 TO 3
    total <- total + scores[i]
ENDFOR
```

```
average <- total DIV 4
OUTPUT average
```

- A. 85
- B. 340
- C. 80
- D. 90

Q27 (3 marks)

What is output by this code?

```
original <- "CODE"
reversed <- ""
FOR i <- LENGTH(original) - 1 DOWNTO 0
    reversed <- reversed + original[i]
ENDFOR
OUTPUT reversed
```

- A. CODE
- B. EDOC
- C. CCCC
- D. OOOO

Q28 (3 marks)

A username must be between 4 and 12 characters (inclusive) and must not be empty. What does this validation function return for `validateUsername("Jo")`?

```
FUNCTION validateUsername(username)
    IF username = "" THEN
        RETURN "Username cannot be empty"
    ENDIF
    IF LENGTH(username) < 4 THEN
        RETURN "Username too short"
    ENDIF
    IF LENGTH(username) > 12 THEN
        RETURN "Username too long"
    ENDIF
    RETURN "Valid"
ENDFUNCTION
```

- ○ **A.** Username cannot be empty
- ○ **B.** Valid
- ○ **C.** Username too long
- ○ **D.** Username too short

Q29 (3 marks)

A cinema offers discounts to people who are either under 16 OR over 65, AND the film must be rated 'U'. Given `age = 70` and `rating = "PG"`, what does this expression evaluate to?

```
(age < 16 OR age > 65) AND rating = "U"
```

- ○ **A.** True
- ○ **B.** PG
- ○ **C.** 70
- ○ **D.** False

Q30 (3 marks)

A programmer is designing a system to manage a library. They identify these main tasks: searching for books, borrowing books, returning books, and managing member accounts. Analyse why decomposing the system into separate subroutines for each task is better than writing all code in one block.

- ○ **A.** Writing all code in one block would be faster to execute
- ○ **B.** Decomposition into subroutines makes the system harder to understand
- ○ **C.** Decomposition offers multiple benefits: each subroutine can be developed and tested independently, different programmers can work on different subroutines simultaneously, bugs are easier to locate within a specific subroutine, subroutines can be modified without affecting others, and code for common operations can be reused across the system
- ○ **D.** One large block of code uses less memory than multiple subroutines

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.